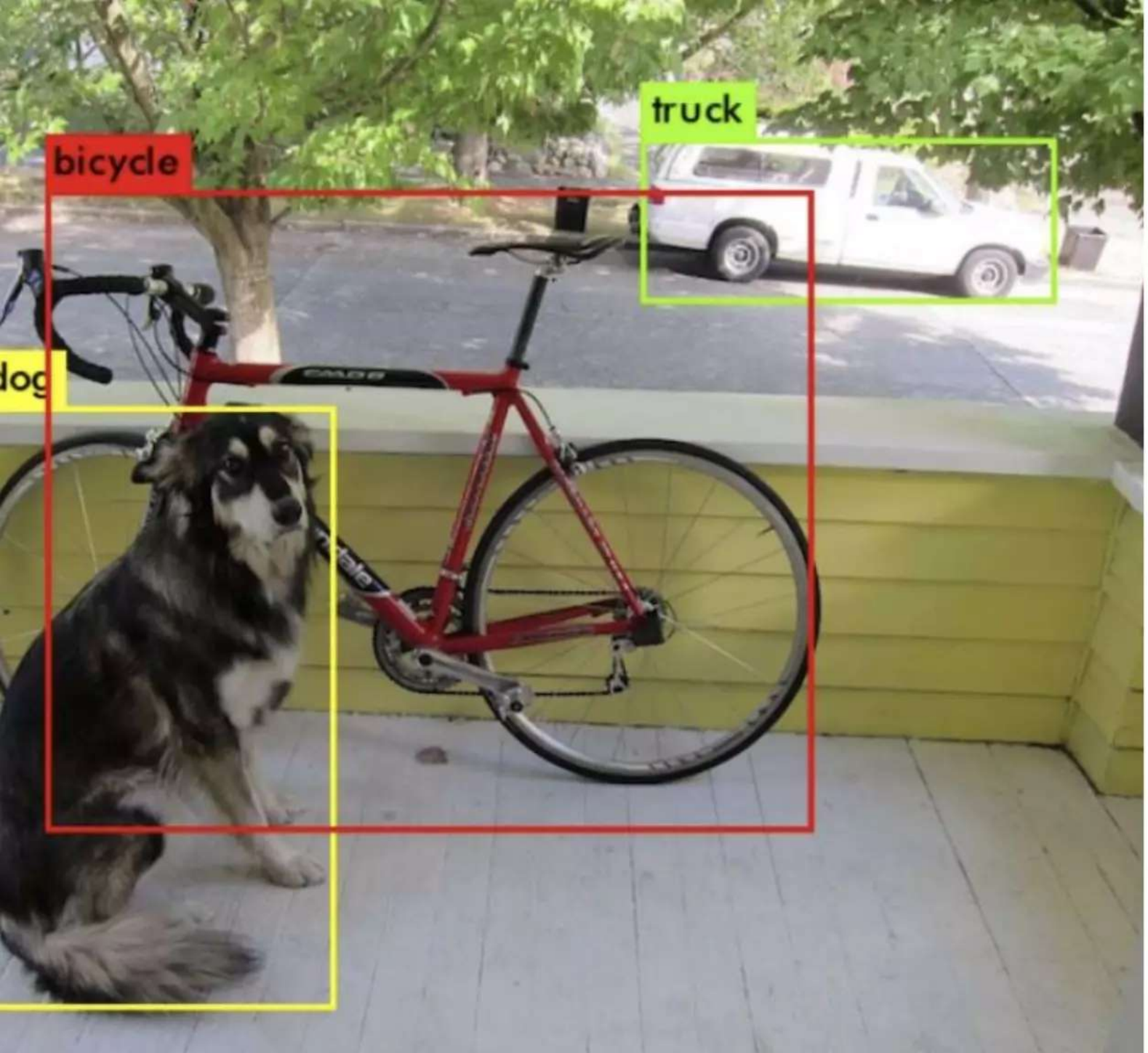

YOLO Algorithm for Object Detection

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Object Detection

It is a phenomenon in computer vision that involves the detection of various objects (eg. people, cars, chairs, stones, buildings, and animals) in digital images or videos.

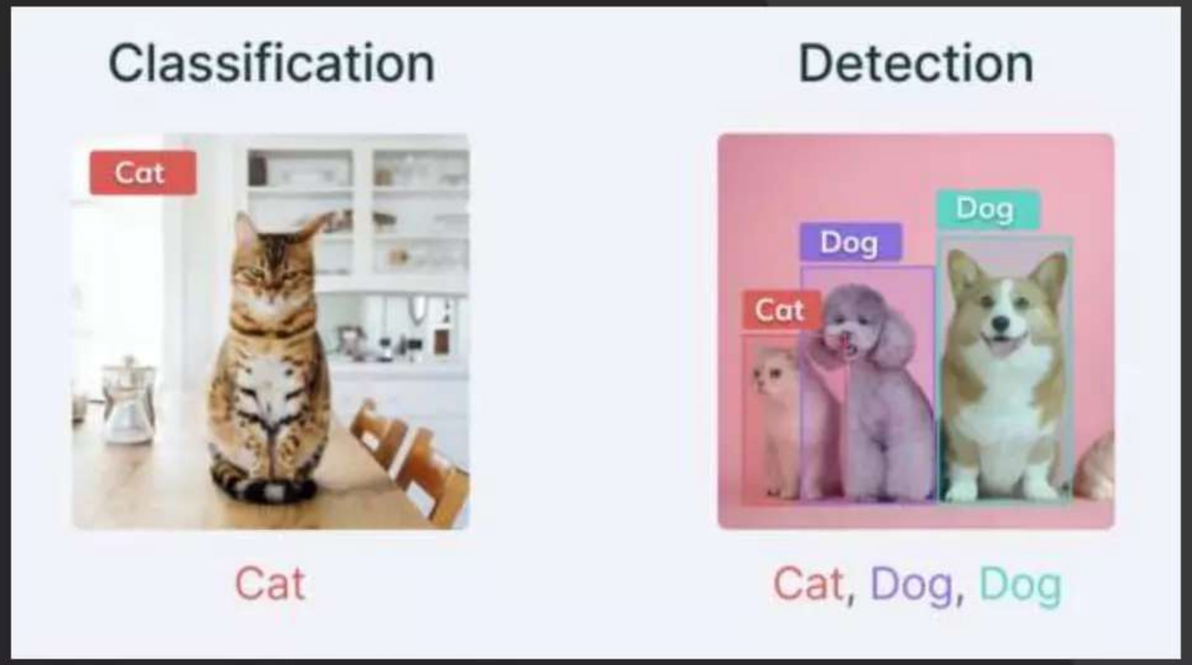
This phenomenon answers two basic questions:

What is the object?

This question seeks to identify the object in a specific image.

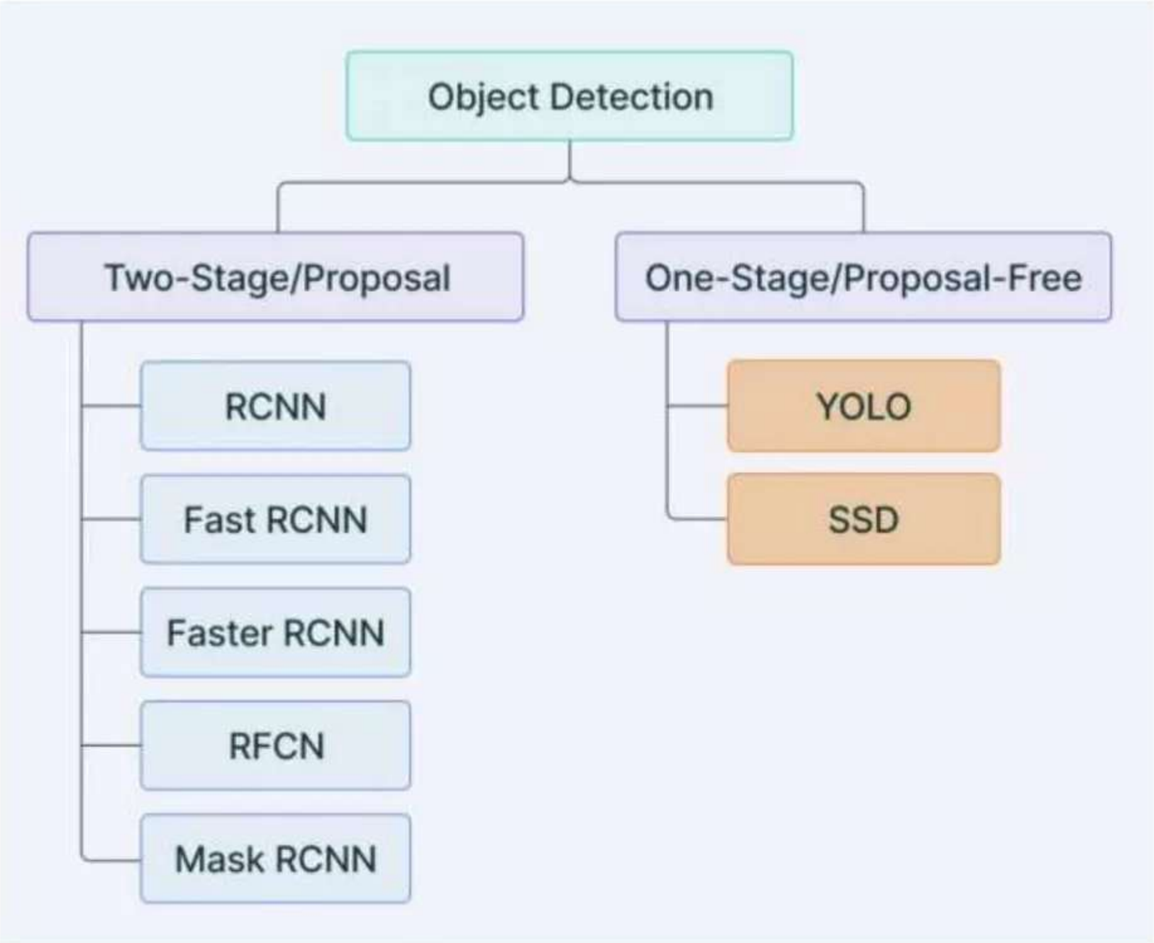
Where is it?

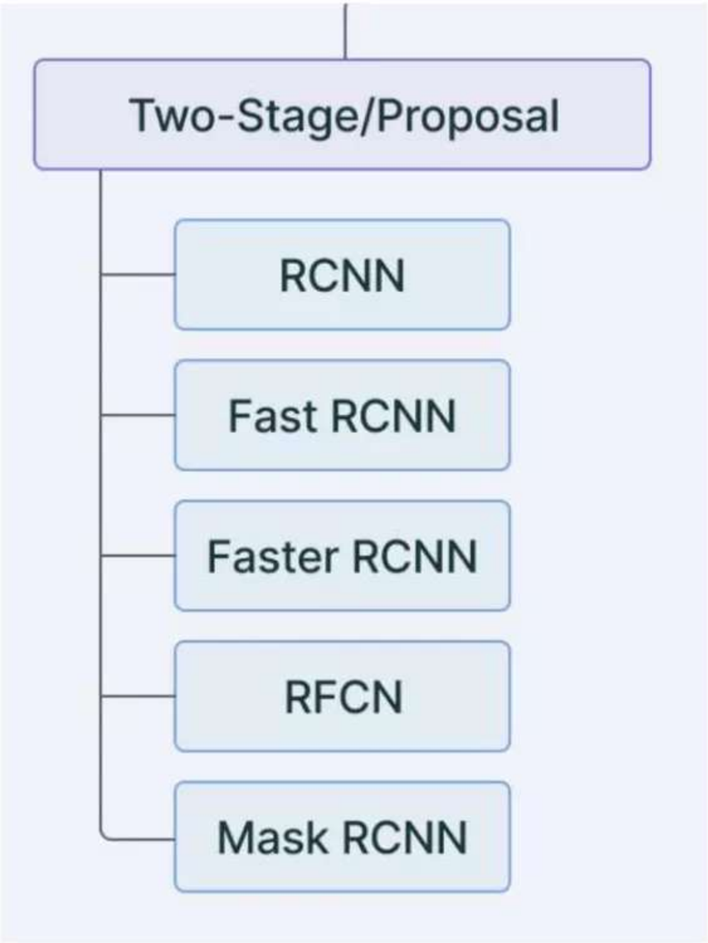
This question seeks to establish the exact location of the object within the image.



Two-stage object detection

- Two-stage object detection refers to the use of algorithms that break down the object detection problem statement into the following two-stages:
- Detecting possible object regions.
- Classifying the image in those regions into object classes.





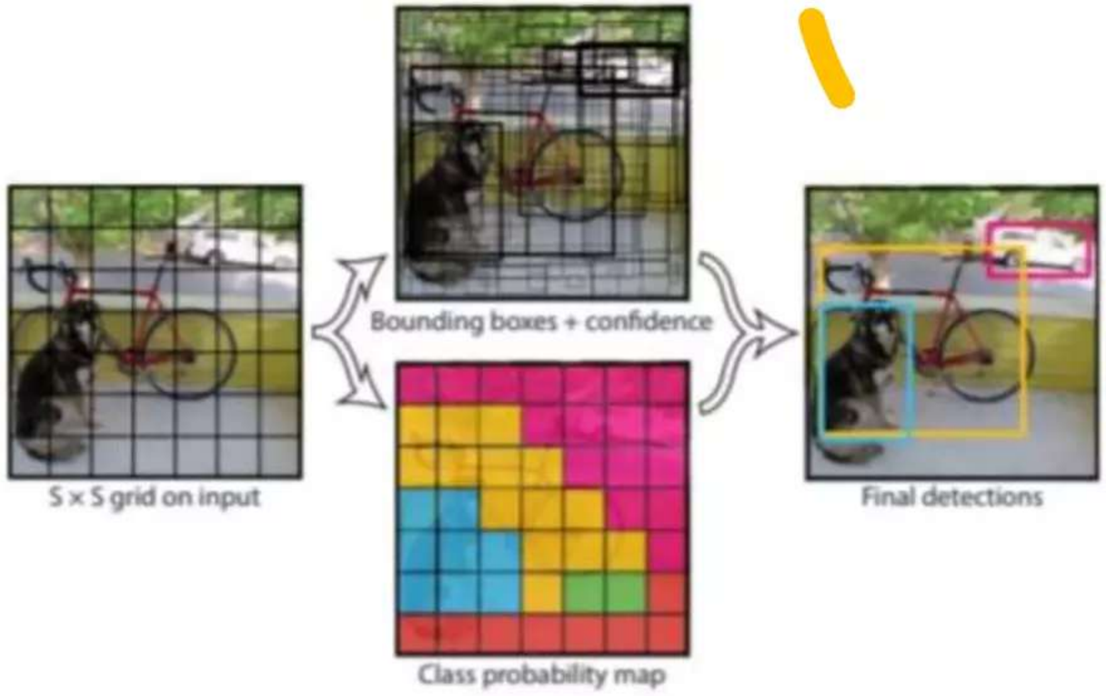
Popular two-step algorithms like Fast-RCNN and Faster-RCNN typically use a Region Proposal Network that proposes regions of interest that might contain objects.

The output from the RPN is then fed to a classifier that classifies the regions into classes.

While this gives accurate results in object detection with a high mean Average Precision (mAP), it results in multiple iterations taking place in the same image, thus slowing down the detection speed of the algorithm and preventing real-time detection.

What is YOLO? You Only Look Once

- YOLO is an algorithm proposed by by Redmond et. al in a research article published at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) as a conference paper, winning OpenCV People's Choice Award.
- In Comparison with other object detection algorithms, YOLO proposes the use of an end-to-end neural network that makes predictions of bounding boxes and class probabilities all at once.



How does YOLO work?

- The YOLO algorithm works by dividing the image into *N* grids, each having an equal dimensional region of SxS.
- Each of these *N* grids is responsible for the detection and localization of the object it contains.
- Correspondingly, these grids predict B bounding box coordinates relative to their cell coordinates, along with the object label and probability of the object being present in the cell.

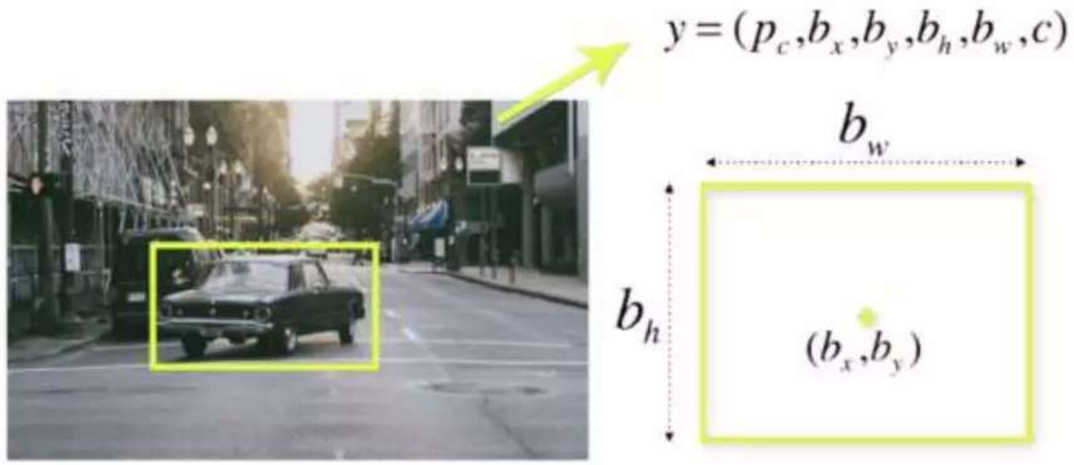
Residual blocks

First, the image is divided into various grids. Each grid has a dimension of S x S. The following image shows how an input image is divided into grids.

ImageSource: https://www.guidetomlandai.com/assets/img/computer_vision/grid.png

Bounding box regression

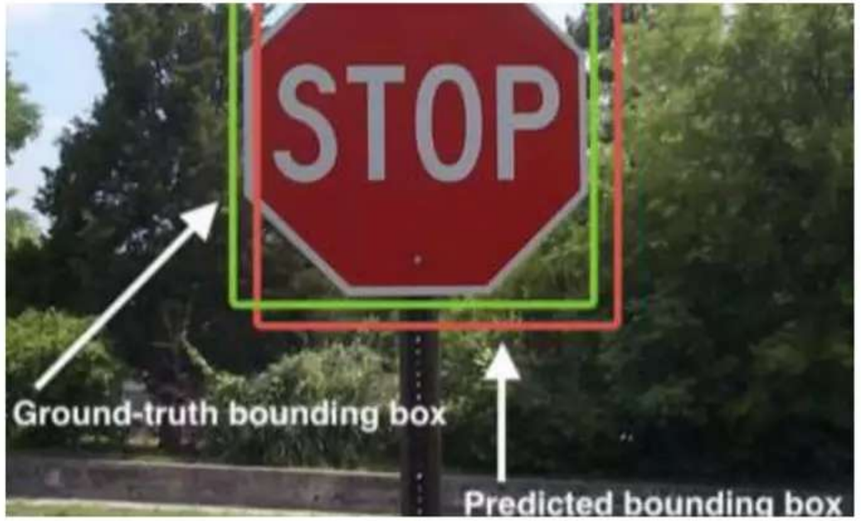
- A bounding box is an outline that highlights an object in an image.
- Every bounding box in the image consists of the following attributes:
- Width (bw)
- Height (bh)
- Class (for example, person, car, traffic light, etc.)- This is represented by the letter c.
- Bounding box center (bx,by)

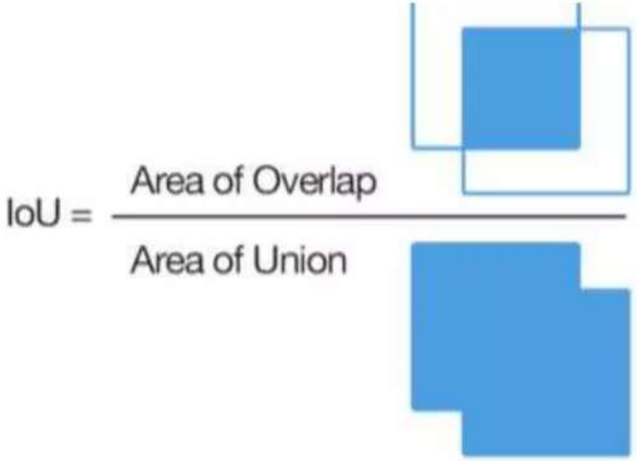


Intersection over Union

- Intersection over Union is a popular metric to measure localization accuracy and calculate localization errors in object detection models.
- To calculate the IoU with the predictions and the ground truth, we first take the intersecting area between the bounding boxes for a particular prediction and the ground truth bounding boxes of the same area. Following this, we calculate the total area covered by the two bounding boxes—also known as the *Union*.
- The intersection divided by the Union, gives us the ratio of the overlap to the total area, providing a good estimate of how close the bounding box is to the original prediction.

Note: Localization error occurs when an object from the target category. is detected with a misaligned bounding box (0.1 <= overlap < 0.5).



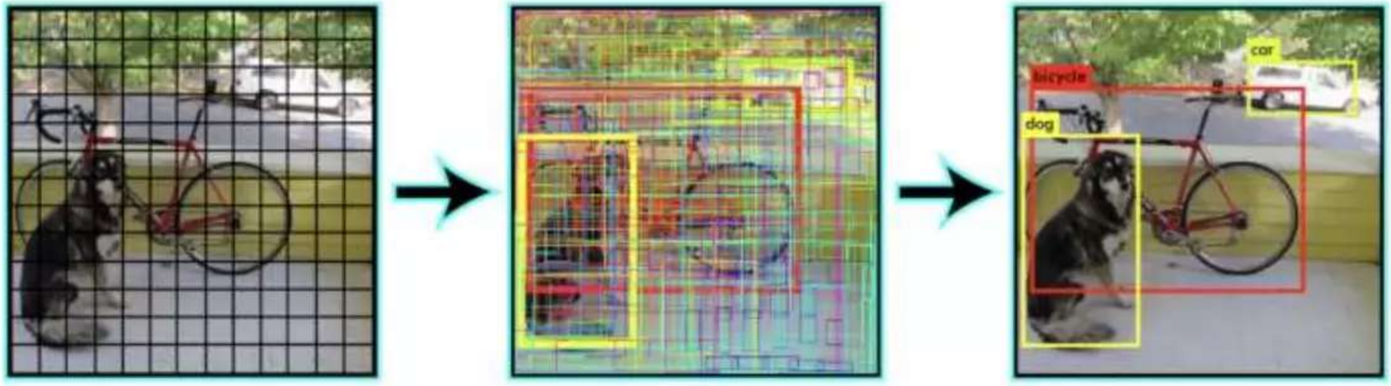


IoU

- Intersection over union ensures that the predicted bounding boxes are equal to the real boxes of the objects.
- This phenomenon eliminates unnecessary bounding boxes that do not meet the characteristics of the objects (like height and width). The final detection will consist of unique bounding boxes that fit the objects perfectly.

Other methods brings forth a lot of duplicate predictions due to multiple cells predicting the same object with different bounding box predictions.

YOLO makes use of Non Maximal Suppression to deal with this issue.



- In Non Maximal Suppression, YOLO suppresses all bounding boxes that have lower probability scores.
- YOLO achieves this by first looking at the probability scores associated with each decision and taking the largest one. Following this, it suppresses the bounding boxes having the largest Intersection over Union with the current high probability bounding box.
- This step is repeated till the final bounding boxes are obtained.

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Average Precision (AP)

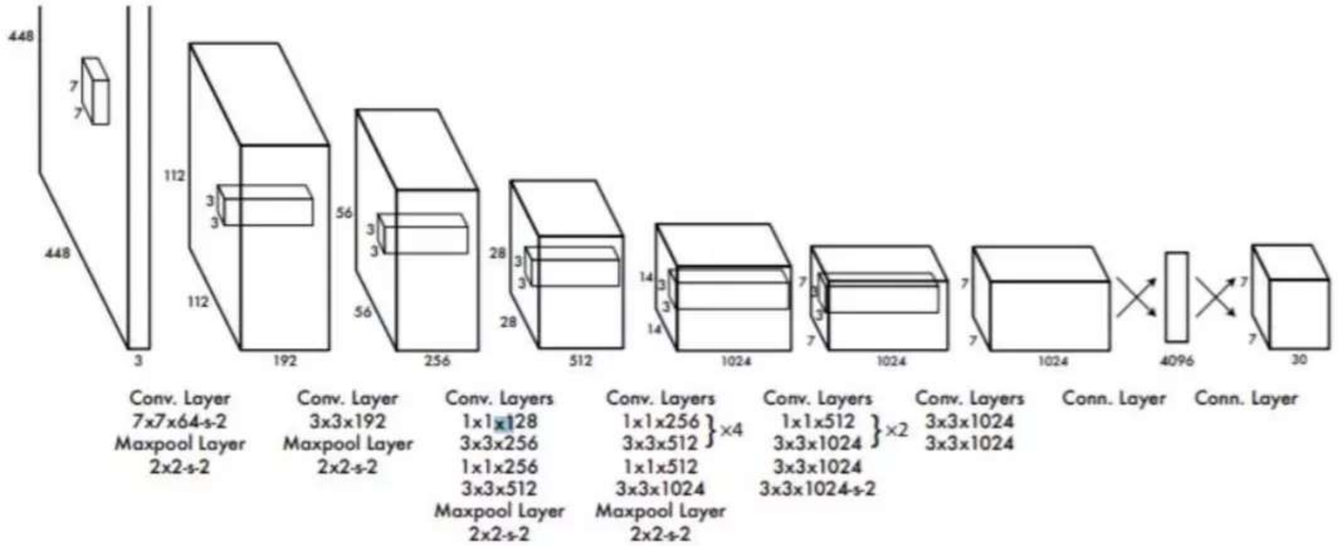
- Average Precision is calculated as the area under a precision vs recall curve for a set of predictions.
- Recall is calculated as the ratio of the total predictions made by the model under a class with a total of existing labels for the class.
- On the other hand, Precision refers to the ratio of true positives with respect to the total predictions made by the model.
- The area under the precision vs recall curve gives us the Average Precision per class for the model. The average of this value, taken over all classes, is termed as mean Average Precision (mAP).

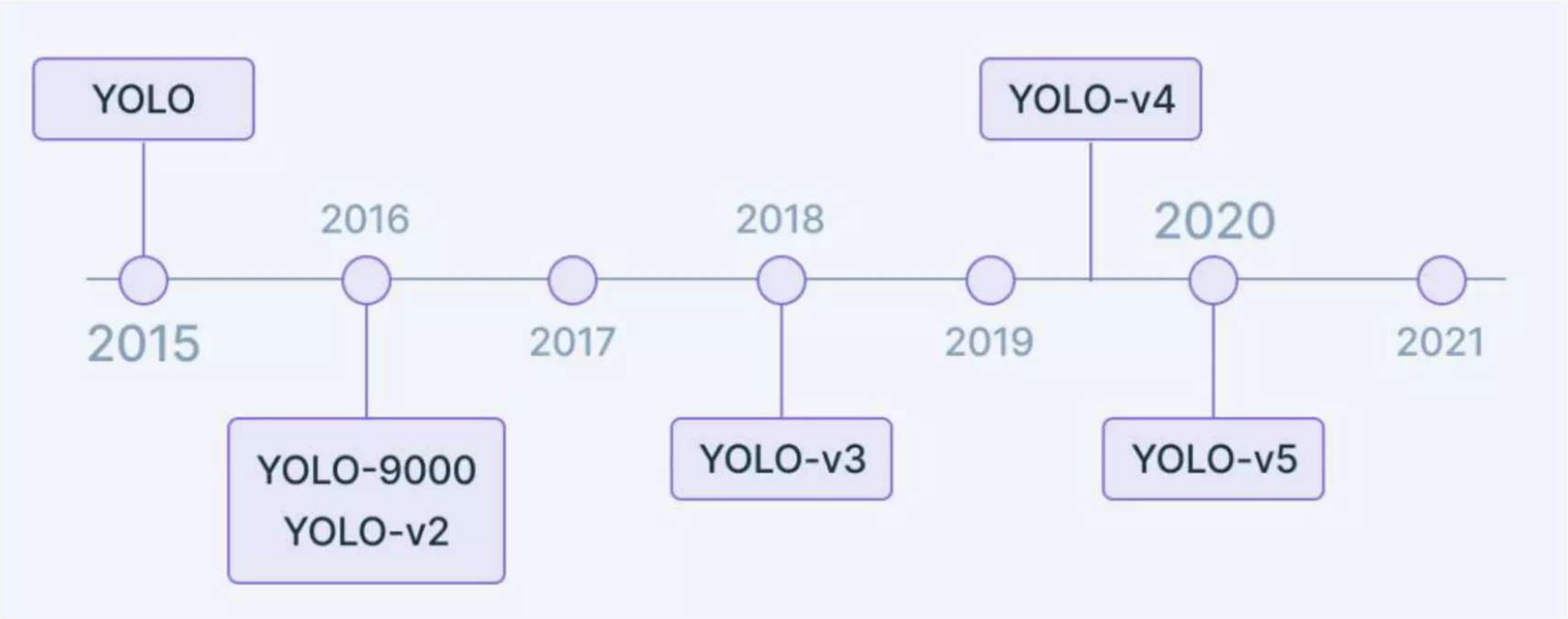
Note: In object detection, precision and recall are not for class predictions, but for predictions of boundary boxes for measuring the decision performance.

An IoU value > 0.5. is taken as a positive prediction, while an IoU value < 0.5 is a negative prediction.

YOLO Architecture

- Inspired by the GoogleNet architecture, The first YOLO architecture has a total of 24 convolutional layers with 2 fully connected layers at the end.





Yolo Timeline

The differences: YOLOv2, YOLO9000, YOLOv3, YOLOv4+

YOLOv2	YOLO9000	YOLOv3	YOLOv4+
YOLOv2 detects even the smallest of objects in groups and the localization accuracy, Anchor boxes	An algorithm to detect more classes than COCO, capable of detecting more than 9000 classes.	YOLOv3 uses a much more complex DarkNet-53 as the model backbone.	Weighted Residual Connections, Cross Mini Batch Normalization
Increased mean Average Precision of the network by introducing *batch normalization*.	However, it provides a lower mean Average Precision as compared to YOLOv2	A 106 layer neural network complete with residual blocks and up-sampling networks	YOLOv5 is an open-source project based on the YOLO model pre-trained on the COCO dataset

Importance of YOLO

- YOLO algorithm is important because of the following reasons:
- **Speed:** This algorithm improves the speed of detection because it can predict objects in real-time.
- **High accuracy:** YOLO is a predictive technique that provides accurate results with minimal background errors.
- **Learning capabilities:** The algorithm has excellent learning capabilities that enable it to learn the representations of objects and apply them in object detection.